



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

Follow For
More Updates



**Class : 12****UNIT TEST-1, JULY - 2024****BUSINESS MATHEMATICS
AND STATISTICS****Time Allowed : 1.30 Hours]****[Max. Marks : 50****Part - I**

1. Answer all the questions by choosing the correct answer from the given 4 alternatives.
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark. 10x1=10

1. If $A = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ then the rank of AA^T is
(a) 0 (b) 1 (c) 2 (d) 3
2. The rank of the unit matrix of order 3 is
(a) 2 (b) 3 (c) 4 (d) 3^2
3. If $T = \begin{matrix} A & \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & x \end{pmatrix} \end{matrix}$ is a transition probability matrix, then the value of x is
(a) 0.2 (b) 0.4 (c) 0.6 (d) 0.8
4. If the rank of the matrix $\begin{pmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ -1 & 0 & \lambda \end{pmatrix}$ is 2. Then λ is
(a) 1 (b) 2 (c) 3 (d) only real number
5. $|A_{n \times n}| = 3$ $|\text{adj}A| = 243$ then the value n is
(a) 4 (b) 5 (c) 6 (d) 7
6. $\int \frac{1}{x^3} dx$ is
(a) $\frac{-3}{x^2} + c$ (b) $\frac{-1}{2x^2} + c$ (c) $\frac{-1}{3x^2} + c$ (d) $\frac{-2}{x^2} + c$
7. $\int \sqrt{e^x} dx$ is
(a) $\sqrt{e^x} + c$ (b) $2\sqrt{e^x} + c$ (c) $\frac{1}{2}\sqrt{e^x} + c$ (d) $\frac{1}{2\sqrt{e^x}} + c$
8. $\int_0^\infty x^4 e^{-x} dx$ is
(a) 12 (b) 4 (c) 4! (d) 64
9. $\Gamma\left(\frac{3}{2}\right)$
(a) $\sqrt{\pi}$ (b) $\frac{\sqrt{\pi}}{2}$ (c) $2\sqrt{\pi}$ (d) $\frac{3}{2}$
10. $\int_0^1 (2x+1) dx$ is
(a) 1 (b) 2 (c) 3 (d) 4

PART - II

1. Answer any 4 questions: 2. Each question carries 2 marks:
3. Question number 16 is compulsory: 4x2=8

11. Find the rank of the matrix $\begin{pmatrix} 1 & -1 \\ 3 & -6 \end{pmatrix}$

12. Show that the equations $x + y = 5$, $2x + y = 8$ are consistent and solve them.

13. Solve the equations $2x + 3y = 7$, $3x + 5y = 9$ by Cramer's rule.

14. Evaluate $\int \frac{dx}{(2x+3)^2}$

15. Evaluate $\int_0^{\infty} e^{-2x} x^5 dx$

16. Evaluate $\int [e^{\frac{1}{2} \log x}]^2 dx$

PART - III

1. Answer any 4 questions. 2. Each question carries 3 marks

3. Question number 22 is compulsory

4x3=12

17. Find the rank of the matrix

$$\begin{pmatrix} -2 & 1 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 1 & 3 & 4 & 7 \end{pmatrix}$$

18. A total of ₹ 8,600 was invested in two accounts. One account earned $4\frac{3}{4}\%$ annual interest and the other earned $6\frac{1}{2}\%$ annual interest. If the total interest for one year was ₹ 431.25, how much was invested in each account? (Use determinant method).

19. Show that the equations $x - 4y + 7z = 14$, $3x + 8y - 2z = 13$, $7x - 8y + 26z = 5$ are inconsistent.

20. Evaluate $\int x^3 e^x dx$

21. $f(x) = \begin{cases} cx, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$ Find 'c' if $\int_0^1 f(x) dx = 2$.

22. Evaluate $\int \frac{4ax+2b}{ax^2+bx+c} dx$

PART - IV

1. Answer all the questions. 2. Each question carries 5 marks

4x5=20

23. a) Solve the following system of equations by rank method $x+y+z=9$, $2x+5y+7z=52$, $2x-y-z=0$.

(OR)

b) Solve by Cramer's rule $x + y + z = 4$, $2x - y + 3z = 1$, $3x + 2y - z = 1$

24. a) Two products A and B currently share the market with shares 50% and 50% each respectively. Each week some brand switching takes place. Of those who bought A the previous week, 60% buy it again whereas 40% switch over to B. Of those who bought B the previous week, 80% buy it again whereas 20% switch over to A. Find their shares after one week and after two weeks. If the price war continues, when is the equilibrium reached?

(OR)

b) Find k, if the equations $x + 2y - 3z = -2$, $3x - y - 2z = 1$, $2x + 3y - 5z = k$ are consistent.

25. a) Evaluate $\int \frac{4x^2+2x+6}{(x+1)^2(x-3)} dx$

(OR)

b) Evaluate $\int (\log x)^2 dx$

26. a) Evaluate $\int_0^{\pi/2} \frac{\sin^7 x}{\sin^7 x + \cos^7 x} dx$

(OR)

b) Evaluate the integral as the limit of a sum: $\int_1^2 (2x+1) dx$



SATGURU STUDY CENTRE

NEVER STOP LEARNING. NEVER STOP GROWING...

17/8, Poondi Thangammal St, Nagoor Garden, Tondiarpet, Chennai, Tamil Nadu 600081

An MSME-Certified Learning Centre

CH/12/Mat/2